In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import (
    confusion_matrix,
    accuracy_score,
    precision_score,
    recall_score,
    f1_score
)
df = pd.read_csv(r"C:/Users/HDC0422234/Downloads/archive (2)/Telco-Customer-Churn.c
print("Dataset Shape:", df.shape)
X = df.drop(['customerID', 'Churn'], axis=1)
y = df['Churn']
print("\nFeatures before encoding:")
print(X.columns.tolist())
print("\nTarget:")
print(y.value_counts())

categorical_columns = X.select_dtypes(
    include=['object']
).columns.tolist()

numerical_columns = X.select_dtypes(
    exclude=['object']
).columns.tolist()
print("\nCategorical Columns:")
print(categorical_columns)
print("\nNumerical Columns:")
print(numerical_columns)
preprocessor = ColumnTransformer(
    transformers=[
        (
            'categorical',
            OneHotEncoder(handle_unknown='ignore'),
            categorical_columns
        )
    ],
    remainder='passthrough'
)
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
print("\nTraining Data:", X_train.shape)
print("Testing Data:", X_test.shape)
logistic_model = Pipeline([
    ('preprocessor', preprocessor),
    ('classifier', LogisticRegression(
```

```
        max_iter=1000,
        random_state=42
    ))
])
decision_tree_model = Pipeline([
    ('preprocessor', preprocessor),
    ('classifier', DecisionTreeClassifier(
        random_state=42
    ))
])
random_forest_model = Pipeline([
    ('preprocessor', preprocessor),
    ('classifier', RandomForestClassifier(
        n_estimators=100,
        random_state=42
    ))
])
print("\nTraining Logistic Regression...")
logistic_model.fit(X_train, y_train)
print("Training Decision Tree...")
decision_tree_model.fit(X_train, y_train)
print("Training Random Forest...")
random_forest_model.fit(X_train, y_train)
print("\nAll models trained successfully!")
y_pred_logistic = logistic_model.predict(X_test)
y_pred_tree = decision_tree_model.predict(X_test)
y_pred_forest = random_forest_model.predict(X_test)
def evaluate_model(model_name, y_test, y_pred):
    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(
        y_test,
        y_pred,
        pos_label='Yes',
        zero_division=0
    )
    recall = recall_score(
        y_test,
        y_pred,
        pos_label='Yes',
        zero_division=0
    )
    f1 = f1_score(
        y_test,
        y_pred,
        pos_label='Yes',
        zero_division=0
    )
    print("\n========================================")
    print(model_name)
    print("========================================")
    print("Accuracy :", round(accuracy, 4))
    print("Precision:", round(precision, 4))
    print("Recall   :", round(recall, 4))
    print("F1-Score :", round(f1, 4))
    cm = confusion_matrix(y_test, y_pred)
    print("\nConfusion Matrix:")
    print(cm)
    return accuracy, precision, recall, f1
logistic_results = evaluate_model(
    "Logistic Regression",
    y_test,
    y_pred_logistic
)
tree_results = evaluate_model(
```

```
    "Decision Tree",
    y_test,
    y_pred_tree
)
forest_results = evaluate_model(
    "Random Forest",
    y_test,
    y_pred_forest
)
results = pd.DataFrame({
    'Model': [
        'Logistic Regression',
        'Decision Tree',
        'Random Forest'
    ],
    'Accuracy': [
        logistic_results[0],
        tree_results[0],
        forest_results[0]
    ],
    'Precision': [
        logistic_results[1],
        tree_results[1],
        forest_results[1]
    ],
    'Recall': [
        logistic_results[2],
        tree_results[2],
        forest_results[2]
    ],
    'F1-Score': [
        logistic_results[3],
        tree_results[3],
        forest_results[3]
    ]
})
print("\n========================================")
print("MODEL PERFORMANCE COMPARISON")
print("========================================")
print(results.round(4))
results.set_index('Model').plot(
    kind='bar',
    figsize=(10, 6)
)
plt.title("Comparison of Classification Models")
plt.xlabel("Models")
plt.ylabel("Score")
plt.ylim(0, 1)
plt.xticks(rotation=0)
plt.legend(title="Metrics")
plt.tight_layout()
plt.show()
results['Average Score'] = results[
    ['Accuracy', 'Precision', 'Recall', 'F1-Score']
].mean(axis=1)
best_model = results.loc[
    results['Average Score'].idxmax(),
    'Model'
]
print("\nBest Performing Model:", best_model)
encoded_features = (
    random_forest_model
    .named_steps['preprocessor']
    .get_feature_names_out()
```

```
)
feature_importance = (
    random_forest_model
    .named_steps['classifier']
    .feature_importances_
)
importance_df = pd.DataFrame({
    'Feature': encoded_features,
    'Importance': feature_importance
})
importance_df = importance_df.sort_values(
    by='Importance',
    ascending=False
)
print("\nTop 10 Factors Influencing Customer Churn:")
print(importance_df.head(10))
top_features = importance_df.head(10).sort_values(
    by='Importance'
)
plt.figure(figsize=(10, 6))

plt.barh(
    top_features['Feature'],
    top_features['Importance']
)
plt.title("Top 10 Factors Influencing Customer Churn")
plt.xlabel("Feature Importance")
plt.ylabel("Features")
plt.tight_layout()
plt.show()
```

```
Dataset Shape: (7043, 21)

Features before encoding:
['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'Mu
ltipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtecti
on', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBillin
g', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges']

Target:
No     5174
Yes    1869
Name: Churn, dtype: int64

Categorical Columns:
['gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines', 'InternetServ
ice', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'Stream
ingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod', 'Total
Charges']

Numerical Columns:
['SeniorCitizen', 'tenure', 'MonthlyCharges']

Training Data: (5634, 19)
Testing Data: (1409, 19)

Training Logistic Regression...
Training Decision Tree...
Training Random Forest...

All models trained successfully!

========================================
Logistic Regression
========================================
Accuracy : 0.7942
Precision: 0.6304
Recall   : 0.5428
F1-Score : 0.5833

Confusion Matrix:
[[916 119]
 [171 203]]

========================================
Decision Tree
========================================
Accuracy : 0.775
Precision: 0.5866
Recall   : 0.516
F1-Score : 0.5491

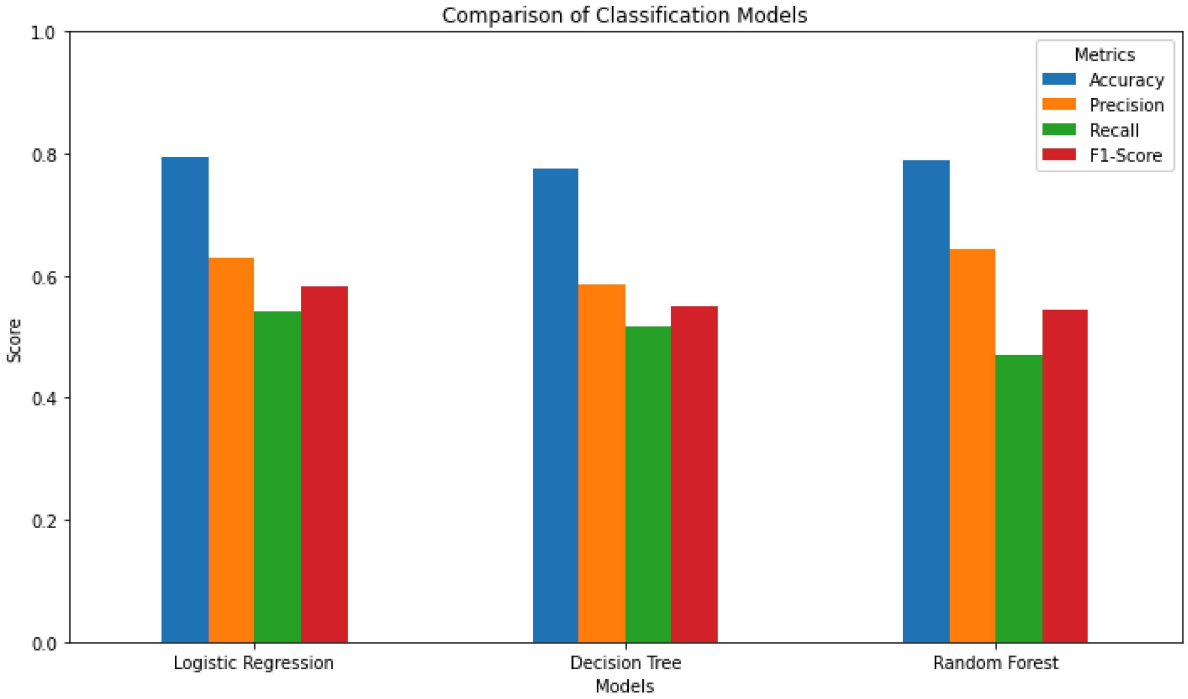
Confusion Matrix:
[[899 136]
 [181 193]]

========================================
Random Forest
========================================
Accuracy : 0.7899
Precision: 0.6423
Recall   : 0.4706
F1-Score : 0.5432

Confusion Matrix:
```

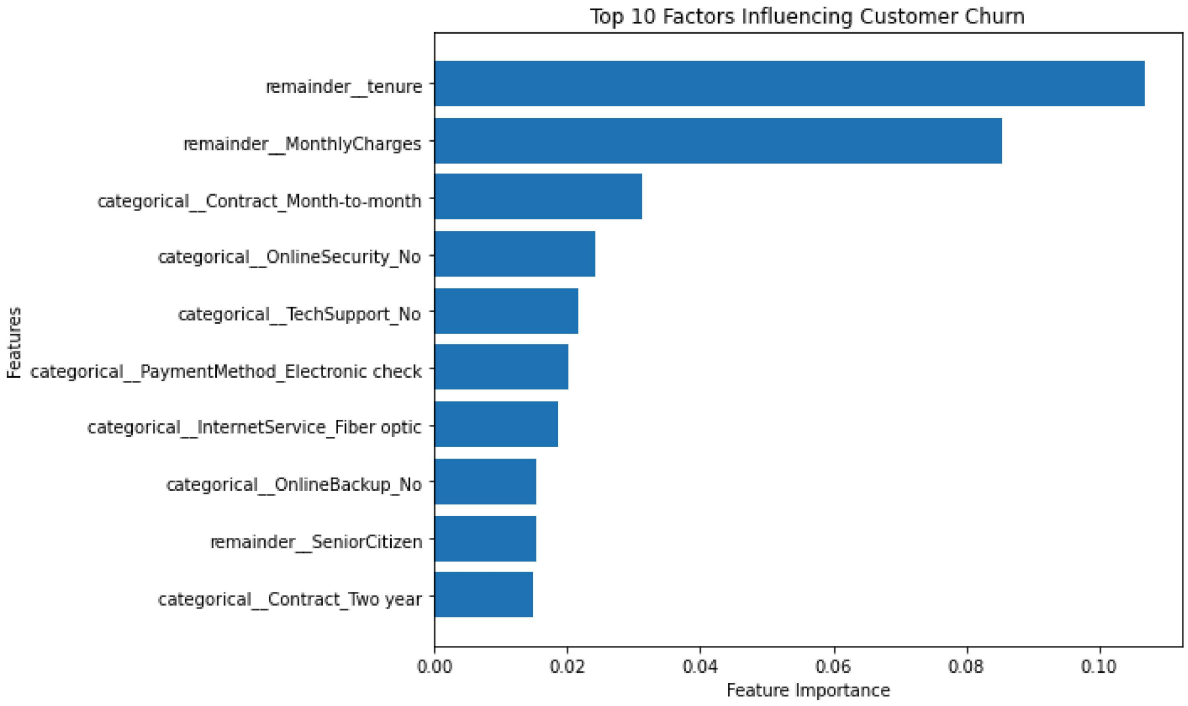
```
[[937  98]
 [198 176]]

========================================
MODEL PERFORMANCE COMPARISON
========================================
                Model  Accuracy  Precision  Recall  F1-Score
0  Logistic Regression    0.7942     0.6304  0.5428    0.5833
1        Decision Tree    0.7750     0.5866  0.5160    0.5491
2        Random Forest    0.7899     0.6423  0.4706    0.5432
```




```
Best Performing Model: Logistic Regression

Top 10 Factors Influencing Customer Churn:
                                        Feature  Importance
5318                          remainder__tenure    0.106955
5319                 remainder__MonthlyCharges    0.085434
32     categorical__Contract_Month-to-month    0.031397
14           categorical__OnlineSecurity_No    0.024160
23              categorical__TechSupport_No    0.021747
39  categorical__PaymentMethod_Electronic check    0.020164
12    categorical__InternetService_Fiber optic    0.018686
17             categorical__OnlineBackup_No    0.015451
5317                remainder__SeniorCitizen    0.015270
34           categorical__Contract_Two year    0.014801
```


Top 10 Factors Influencing Customer Churn
remainder__tenure
remainder__MonthlyCharges
categorical__Contract_Month-to-month
categorical__OnlineSecurity_No
categorical__TechSupport_No
categorical__PaymentMethod_Electronic check
categorical__InternetService_Fiber optic
categorical__OnlineBackup_No
remainder__SeniorCitizen
categorical__Contract_Two year
Features
0.00
0.02
0.04
0.06
0.08
0.10
Feature Importance

In [ ]: